Vel Tech Group of Institutions

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2028_NeoColab_Competitive Coding II_L8

VTU_Week 8_MCQ

Attempt : 1
Total Mark : 15
Marks Obtained : 10

Section 1 : MCQ

1. Which data structure is used in Breadth First Search?

Answer

Queue

Status : Correct ***Marks : 1/1***

2. Which of the following problems can be solved by using both BFS & DFS?

Answer

Finding the shortest path between two vertices in an unweighted graph

Status : Correct ***Marks : 1/1***

3. In BFS, what happens when a node is visited for the second time?

Answer

It is skipped and the traversal continues to the next unvisited node

Status : Correct ***Marks : 1/1***

4. Given two vertices in a graph s and t, which of the two traversals (BFS and DFS) can be used to find if there is path from s to t?

Answer

Both BFS and DFS

Status : Correct ***Marks : 1/1***

5. What will be the result of depth-first traversal in the following tree?

Answer

1 2 4 5 3

Status : Correct ***Marks : 1/1***

6. The Breadth First Search (BFS) algorithm has been implemented using the queue data structure. Which one of the following is a possible order of visiting the nodes in the graph below?

Answer

NQMPOR

Status : Wrong ***Marks : 0/1***

7. In the _______ traversal, we process all of a vertex's descendants before we move to an adjacent vertex.

Answer

Depth First

Status : Correct *Marks : 1/1*

8. Which of the following conditions is sufficient to detect a cycle in a directed graph?

Answer

There is an edge from the currently visited node to an ancestor of the currently visited node in the DFS forest.

Status : Correct *Marks : 1/1*

9. Regarding the implementation of Breadth First Search using queues, what is the maximum distance between two nodes present in the queue? (considering each edge length 1)

Answer

At most 1

Status : Correct *Marks : 1/1*

10. Given the following adjacency matrix of a graph(G) determine the number of components in the G.

[0 1 1 0 0 0],

[1 0 1 0 0 0],

[1 1 0 0 0 0],

[0 0 0 0 1 0],

[0 0 0 1 0 0],

[0 0 0 0 0 0].

Answer

3

Status : Correct *Marks : 1/1*

11. Which of the following statements is true about DFS?

Answer

DFS always finds the shortest path between two nodes in a graph.

Status : Wrong ***Marks : 0/1***

12. Consider the following graph.

Apply breadth-first search traversal of the above graph. Which of the following traversal is possible if the start vertex is G? (Assume lexicographic ordering)

Answer

GAHIJBCEKD

Status : Wrong ***Marks : 0/1***

13. Which of the following is a possible result of depth-first traversal of the given graph(consider 1 to be the source element)?

Answer

1 4 5 3 2

Status : Wrong ***Marks : 0/1***

14. In an adjacency list representation of a graph, what does each node in the list represent?

Answer

Edges

Status : Wrong ***Marks : 0/1***

15. The number of elements in the adjacency matrix of a graph having 7

vertices is _________.

Answer

49

Status : Correct ***Marks : 1/1***